SINGAPORE SCHOOL KINDERLAND

Department of Information Technology & Computer Science

ACADEMIC YEAR 2026–2027 · TERM 2 MASTER SYLLABUS OUTLINE

Class / Grade	Term 2 Chapters & Book Pages	Core Topics & Learning Scope	Status
Primary 1	**Chapter 4: Parts of a Computer (Pg 26–33)** **Chapter 5: Introduction to Logic (Pg 34–44)**	• Ch 4: Desktop parts (CPU, Monitor, Keyboard, Mouse, Printer, Speakers, Mic, Webcam), Laptop components (Screen, Touchpad, Camera), Portable vs Non-portable. • Ch 5: Actions and Reasons, Logical everyday pairs (Coffee/Cup, Cricket/Bat, Rain/Raincoat), Categories (Fruits, Furniture, Vehicles), Odd one out.	**Finalized (Book Verified)**
Primary 2	**Ch 2B: Keyboard & Notepad (Pg 18–26)** **Ch 3: Windows 10 O/S (Pg 27–38)** **Ch 4: Algorithm (Pg 39–48)**	• Ch 2B: Alphabet, Number & Special keys (Spacebar, Enter, Backspace, Delete, Caps Lock, Shift); Notepad opening, typing & saving (.txt). • Ch 3: O/S concept, Desktop, Taskbar, Start Menu, Icons, Wallpaper, Screen Saver, File & Folder management. • Ch 4: Algorithm definition, Daily life step sequencing (Making tea, Brushing teeth, Packing bag), Computer problem solving.	**Finalized (Book Verified)**
Primary 3	**Ch 3: Internet & Networks (Pg 18–28)** **Ch 4: E-Commerce (Pg 29–38)** **Ch 5A: MS Word Formatting (Pg 39–50)**	• Ch 3: Network types (LAN, MAN, WAN), WWW, Webpages, Websites, URLs, Web Browsers vs Search Engines, Netiquette & Cyber safety. • Ch 4: E-Commerce advantages, Models (B2C, B2B, C2C), Payment methods (Cards, Mobile wallets, COD), Security (OTP, Passwords). • Ch 5A: MS Word interface (Ribbon, Quick Access Toolbar), Editing (Cut, Copy, Paste, Find/Replace), Formatting (Font, Alignment, Lists).	**Finalized (Book Verified)**
Primary 4	**Ch 3: Internet & Networks (Pg 18–28)** **Ch 4: E-Commerce (Pg 29–38)** **Ch 5A: MS Word Formatting (Pg 39–50)**	• Ch 3: Computer Networks & Architecture, Internet navigation, Web Browsers, Hyperlinks, Search tips & Netiquette. • Ch 4: Online business concepts, Digital marketplaces, Electronic payments & Security best practices. • Ch 5A: Word processing workflows, Text formatting, Alignment styles, Bulleted lists, Document layout & Proofreading. *(Curriculum aligned with Primary 3 track)*	**Finalized (Shared Track)**
Primary 6	**Ch 2: OpenOffice I - Writer & Impress (Pg 12–28)** **Ch 3: OpenOffice II - Calc & Base (Pg 29–48)**	• Ch 2: Open Source Software; Writer (Interface, Margins, Spacing, Tables, Images, Header/Footer); Impress (Presentation slides, Transitions, Custom Animations, Slide show). • Ch 3: Calc (Spreadsheet grid, Cell referencing, Formulas, Functions: SUM, AVERAGE, COUNT, MAX, MIN, Charts); Base (DBMS concepts, Tables, Records, Fields, Primary Key, Queries, Reports).	**Finalized (Book Verified)**
Secondary 1 (Sec 1)	**Chapter 2: Data transmission (Pg 54–74)** **• 2.2 Error detection (Pg 54–62)** **• 2.3 Encryption (Pg 63–74)**	• 2.2 Methods of Error Detection: Parity checks (even/odd parity bits, parity block), Checksum calculation, Echo check, Check digits (ISBN 13, Modulo 11), Automatic Repeat Requests (ARQ) with timeouts & acknowledgements. • 2.3 Symmetric and Asymmetric Encryption: Symmetric encryption (shared secret key, plain text to cipher text), Asymmetric encryption (public & private key pairs), Hashing, Digital certificates (SSL/TLS) & brute force attacks.	**Finalized (Cambridge 0478)**
Secondary 2 (Sec 2)	**Chapter 1: Data representation (Pg 32–44)** **• 1.3 Data storage & compression (Pg 32–44)** **Chapter 2: Data transmission (Pg 54–74)** **• 2.2 Error detection & 2.3 Encryption (Pg 54–74)**	• 1.3 Data Storage & Compression: Storage measurement (bits, bytes, KiB/MiB/GiB vs KB/MB/GB), File size calculations (sound, image & text files), Lossy vs Lossless compression, Run-length encoding (RLE). • 2.2 Error Detection: Parity bits & blocks, Checksum, Echo check, Check digits (Modulo 11), ARQ protocol. • 2.3 Encryption: Symmetric vs Asymmetric encryption, Public/Private key pairs, Plain/Cipher text, Digital signatures.	**Finalized (Cambridge 0478)**

Academic Reference & Course Mapping: Primary 1–6 curricula follow approved textbook editions. Secondary 1 and Secondary 2 curriculum strictly adheres to *Cambridge IGCSE & O Level Computer Science (0478/2210) Second Edition* (Ch 1.3 Data Storage/Compression & Ch 2 Data Transmission: Error Detection 2.2 & Encryption 2.3).